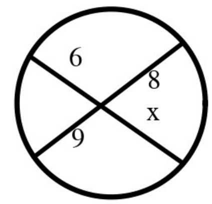
Speak **L**ike **A** **M**athematician:

<u>segments of a chord:</u>

<u>secant segment / external segment:</u>



<u>Segment Product Rules</u>

Two chords: (piece)(piece) = (____________)(____________) of the other chord.

Two secants: (whole)(____________________) = (whole)(external) of the other secant.

Tangent and secant: tangent² = (____________)(____________________).

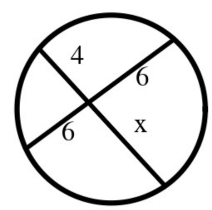
Example 1 – Two Chords

In the figure at the top right, the pieces are 6 and x on one chord, 8 and 9 on the other.

a) Find x.

Example 2 – Two Chords Again

In the figure at the right, the pieces are 4 and x on one chord, 6 and 6 on the other.

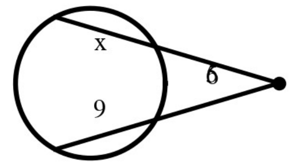


a) Find x.

Example 3 – Two Secants

In the top figure, one secant has external part 5 and far part x; the other has external part 6 and far part 9.

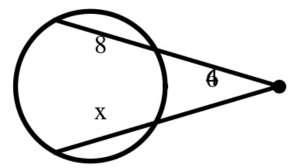
a) Find x.



Example 4 – Two Secants Again

In the middle figure, external 4 with far part 8; external 6 with far part x.

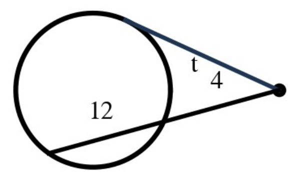
a) Find x.



Example 5 – Tangent and Secant

In the bottom figure, the tangent is t; the secant has external part 4 and far part 12.

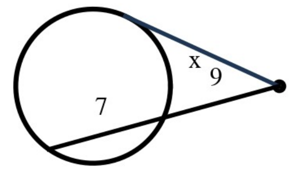
a) Find t.



Example 6 – Tangent and Secant Again

In the top figure, tangent x; secant external 9, far part 7.

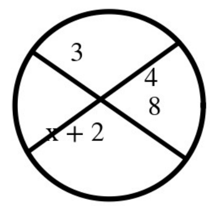
a) Find x.



Example 7 – Chords with an Expression

In the middle figure, pieces 3 and 8 on one chord; 4 and (x + 2) on the other.

a) Find x.



Example 8 – Tangent, Solving a Quadratic-Free Way

In the bottom figure, tangent 10; secant external 5, far part x.

a) Find x.

b) What is the whole secant length?